Proceedings of Machine Learning Research vol vvv:1–12, 2027

The Guarantee Holds and the Task Fails: Absorbing Stalls of Learned Policies under CBF Filters

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Abstract

A safety filter that projects a learned policy's command onto an admissible set guarantees forward invariance of the safe set, and nothing about progress. We show that a control barrier function (CBF) filter in front of a memoryless learned pursuer creates a closed-loop fixed point at which the guarantee holds, violations are zero, and the task fails silently: the filter zeroes the command, the body stops, the observation freezes, and the policy re-emits the same command for the rest of the episode. We state the fixed point as the normal-cone condition for CBF-QP deadlock applied to a learned nominal and measure it: 98% of stalled states are traps at the current heading, none is a trap of the position. We measure why learning does not remove it: the reward for escaping arrives 629 steps later and is worth 0.009 at $\gamma = 0.99$. Across seven pre-registered rounds, every learning-side remedy improves the task and none removes the failure, a stall penalty makes it worse, and a one-counter recovery that offers filtered forward and rotation candidates after N immobile steps clears the deadlock on three of three policy families (0.19–0.30 to 0.000), raises capture on nine of nine seeds with zero violations, is insensitive to N over 5–80 steps, and composes onto a second filter. Three dissociations show liveness metrics improving while the task collapses.

Keywords: safety filters; control barrier functions; shielding; safe reinforcement learning; deadlock; liveness; pursuit–evasion

1. Introduction

Safety filters wrap a learned policy with a guarantee. A control barrier function (CBF) filter, a shield, or a predictive filter takes the policy's command, tests it against a safe set, and replaces it with the nearest admissible command when the test fails (Hsu et al., 2024; Ames et al., 2019; Alshiekh et al., 2018; Cheng et al., 2019). The guarantee is forward invariance of the safe set; it says nothing about the task, and the two can come apart in a way no violation count reveals.

Figure 1 shows the failure in one archived episode. A PPO-trained pursuer behind a CBF filter drives toward an obstacle between it and the evader; the filter zeroes its forward command. The body stops, the observation stops changing, and the memoryless policy emits the same full-speed command at each of the next 1 106 steps (standard deviation 0.000) while the executed speed stays at zero. The episode ends in a timeout. The same episode under the recovery of Section 4, same seed and same evader, ends in a capture at step 262. The violation count is zero in both.

This is not the multi-robot deadlock of the CBF literature, a force equilibrium between agents whose constraint normals cancel the nominal (Grover et al., 2020, 2021, 2023); here one agent stands against static geometry. The nearest prior objects are the undesired equilibria that CBF quadratic programs introduce into the closed loop of a fixed nominal controller (Reis et al., 2021; Tan and Dimarogonas, 2024). Ours is a nominal trained through the filter that sees the world only through an observation the stall has frozen: the trap is in observation space, and that is why learning does not escape it.

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Contributions. (C1) A characterisation of filter-induced stalls for learned memoryless policies, separating a trap of the position from a trap of the policy–filter pair at a heading, measured on 262 stalled states (Proposition 1, round B1). (C2) The mechanism that defeats learning: a frozen input and a reward for escape that arrives about 630 steps later (round S15). (C3) A pre-registered comparison in which every learning-side remedy improves the task and none removes the failure, a stall penalty backfires, and a one-counter forward-offering recovery removes the failure with the guarantee untouched, across three policy families, a parameter sweep, a perturbation baseline, and a second filter class (Proposition 3, rounds S16, S17, B2–B4). (C4) Three pre-registered dissociations between filter-adjacent liveness metrics and the task (rounds S16, S18, S19).

2. Related work

Deadlock and undesired equilibria of CBF-QP controllers. Grover et al. (2020, 2021, 2023) characterise multi-robot deadlock under CBF quadratic programs through the Karush–Kuhn–Tucker conditions: the projected command is zero exactly when the nominal lies in the conic hull of the active constraint normals. Reis et al. (2021) showed that CLF–CBF programs create spurious stable equilibria on the unsafe set's boundary for a single agent and one obstacle; Tan and Dimarogonas (2024) characterised all closed-loop equilibria and modified the program to remove the undesired ones while keeping forward invariance; Reis and Aguiar (2026); Reis et al. (2025) extended the analysis, the latter noting that liveness is not guaranteed. These works fix the filter; we keep it and its guarantee untouched, show that a learned nominal makes the equilibrium absorbing in observation space, and remove it with a stateful shield outside the filter.

Shield conservativeness. Kim et al. (2025) require that a continuous-space shield always has a safe action to offer, including stateful requirements such as loop avoidance, and note that the shield alone is overly conservative; Dawood et al. (2025) adapt an MPC shield's weights with a learned supervisor; Wang et al. (2025) resolve multi-robot deadlocks after they occur; Arnström and Teixeira (2025) study adversarial filter deactivation. Our remedy is one counter, evaluated on learned policies in an adversarial game; Proposition 1 predicts the oscillations and timeouts these papers report.

Learning through the filter. Oh et al. (2026) show that a filter-unaware policy chatters against the filter and that training through it is optimal when the filter is permissive enough; Wu et al. (2026) train through a filter for minimally exploitable game policies; Georgescu et al. (2026) repair a shield's specification online to keep liveness achievable; Pizarro Bejarano et al. (2025) penalise the filter's correction during training. We measure the case in which training through the filter still fails: at the stall the filter is not permissive, and the signal that would teach the escape is discounted away. We found no prior work that separates filter-adjacent liveness metrics from task success under pre-registration (Section 6).

3. Setting and definitions

Game. A unicycle pursuer ($v \le 1\,\mathrm{m/s}$, $|\omega| \le 2.84\,\mathrm{rad/s}$, $\Delta t = 0.05\,\mathrm{s}$) hunts an evader in a $10 \times 10\,\mathrm{m}$ arena with two circular obstacles, sensing within a 90°, $8\,\mathrm{m}$ field of view; capture is contact within $0.30\,\mathrm{m}$; episodes last 1 200 steps. The evaders are a greedy flee policy, an obstacle-hugging policy, and a learned full-state attacker (ATK4-45) trained to stall filtered pursuers. The

Table 1: Pre-registered rounds: frozen commit, hypotheses and bars fixed before any number, paired seeds on a fresh stream, records under version control. H-SAFE held (zero contact steps) in every filtered cell of every round.

Round	Measures	Policies	Opponents	N / cell
S4, S5	filter cost; first remedies	STR14; RA, RB, RC	six	500 paired
S15	mechanism (772 stall entries)	FT-B-41, RC19	ATK4-45, Greedy	200
S16	five learned remedies; rotation recovery	BASE, PEN, MEM, OBS, GAM, REC	ATK4-45, HUG7, Greedy	500 paired
S17	forward-offering recovery, seven families	BASE, OBS, GAM, OBSGAM, RECF	ATK4-45, HUG7, Greedy	500 paired
S18	correction-magnitude penalty, 18 arms	BASE, DVARY, SPARSE, DRARE	Greedy	500
S19	law pins removed (planner law)	LINE, PLAN	Greedy, HUG, ATK4-45	500 paired
B1	stall geometry; dwell (det. / stoch.)	BASE-60/61/62, RC19	ATK4-45	100
B2–B4	N sweep; baseline; second filter (86 cells)	BASE-60/61/62, STR14, RC19	Greedy, ATK4-45	500 paired

pursuers are structured search policies: a PPO head whose search action is held for 10 steps while the evader is out of view, and a fixed intercept law when it is visible; both are memoryless.

Filter. The base filter is a discrete-time CBF projection: a command is admissible if the exact one-step barrier condition holds on the true Euler successor, and the filter executes the admissible command nearest to the nominal in the weighted norm $w_v(\Delta v)^2 + w_\omega(\Delta\omega)^2$ ($w_v = 150$, $w_\omega = 1$), with barrier rate 0.2 and a virtual control point $0.1\,\mathrm{m}$ ahead of the axle; it has exact, relaxed and backup tiers, and every stalled state in this paper lies on the exact tier. The second filter (Section 5.3) is a sampling lookahead through the simulator with horizon 4. A violation is the simulator's contact flag; H-SAFE requires zero contact steps in every filtered cell.

Definitions. A step is *immobile* if the executed speed is at most $0.05\,v_{\max}$; *stalled* if immobile while the nominal speed is at least $0.5\,v_{\max}$; a *self-stop* if immobile and not stalled. A *stall run* is a maximal sequence of stalled steps; a *deadlock episode* contains a stall run of at least $K = 40$ steps ($2\,\mathrm{s}$). A state is a *filter trap at heading* if no admissible command at the current pose moves the body (speed above $0.05\,v_{\max}$), a *position trap* if this holds at every heading there, and a *pair trap* if it is the former and not the latter. An *escape* is a displacement of $0.10\,\mathrm{m}$ from the stalled pose; an *engagement* is a cluster of recovery firings fewer than five steps apart; a *cycle* is an engagement that ends in a stall within 40 steps of the previous one.

Policies and rounds. Table 1 lists the rounds. The BASE, OBS, GAM and OBSGAM families (three seeds each) are trained from scratch for 5 M steps behind the filter; OBS adds the filter's admissible-turn features to the observation, GAM raises the discount to 0.999, OBSGAM does both. STR14 and RC19 are 10 M-step policies from the rounds in which the failure was first seen; RC19 observes the filter's state. All policies are frozen before evaluation.

4. Why a filtered memoryless policy stops

Let s be the pursuer's state, $o(s)$ its observation, $u_\pi = \pi(o(s))$ the nominal command, and $\mathcal{A}(s) \subset [0, v_{\max}] \times [-\omega_{\max}, \omega_{\max}]$ the set of commands that pass the exact one-step barrier test at s: the command box intersected with the half-spaces of the linearised barrier rows, hence convex. On the exact tier the filter executes $P_{\mathcal{A}(s)}(u_\pi)$, the unique projection of the nominal onto $\mathcal{A}(s)$ in the weighted norm.

Proposition 1 (Stall characterisation) *On the exact tier the executed speed is zero if and only if u_π lies in the weighted normal cone of $\mathcal{A}(s)$ at the zero-speed face, equivalently $u_\pi = \sum_{j \in J(s)} \mu_j a_j$ with $\mu_j \geq 0$ over the active barrier normals a_j. If moreover $o(s)$ is constant along the stall, the closed loop is at a fixed point and the stall is absorbing for the rest of the episode.*

Proof [Proof sketch] Projection onto a closed convex set returns the $u^\star$ with $\langle u_\pi - u^\star, u - u^\star \rangle_W \leq 0$ for all admissible u; on the zero-speed face this is the normal-cone condition, and stationarity of the Lagrangian gives the conic combination of active normals, the force-equilibrium condition of Grover et al. (2023) for one agent and static constraints. If $o(s)$ is constant, so are u_π, the projection, and s. ■

The characterisation is known; the learned setting adds four refinements, measured in Section 5. (R1) *Position trap versus pair trap.* A stall at pose (p, θ) can be a trap at heading θ while another heading at p admits forward motion; what is trapped is the pair (policy, filter): the policy asks for the blocked direction, and the filter admits only commands the policy never asks for. (R2) *Eventual constancy.* The observation depends on the pose, the opponent's visibility, and bounded filter features whose stall counter saturates at 80 steps, so with the opponent out of view the input is constant after at most 80 stalled steps; filter-state features (the OBS family, RC19) therefore move the fixed point without removing it. (R3) *Exploration.* One might expect a Gaussian head's stall to end once a sampled action displaces the body, with a geometric dwell. Round B1 does not support this: the long stalls belong to the intercept law, which samples nothing, and where the head is in control the evader's motion ends the stall sooner than sampling would; what survives is that deadlock under stochastic sampling is no rarer than under deterministic actions. (R4) *Two kinds of zero.* The nominal may itself be zero; a policy that has learned not to ask reaches the same fixed point as one whose request is blocked (Section 6).

Remark 2 (Why learning does not remove it) *Let A_0 be the undiscounted capture advantage of escaping a stall and T the delay from the escape to the capture. By (R2) the advantage attaches to one shared input vector, and under the training discount it is at most $\gamma^T A_0$. Round S15 measured $A_0 = 0.257$ and $T \approx 629$ steps: 0.0005 at $\gamma = 0.99$ (measured discounted advantage 0.0092) and 0.137 at $\gamma = 0.999$. This bounds a learning signal; it does not prove failure. The GAM family ($\gamma = 0.999$) is the test: it raises the task and does not clear the deadlock.*

The recovery. After N consecutive immobile steps, the shield filters the candidates $(v_{\max}, 0)$, $(\frac{1}{2} v_{\max}, 0)$, the nominal, and $(0, \pm\omega_{\max})$ in that order and executes the first whose filtered speed exceeds $0.05\, v_{\max}$; if none moves, it executes the larger admissible rotation. The counter resets on any mobile step.

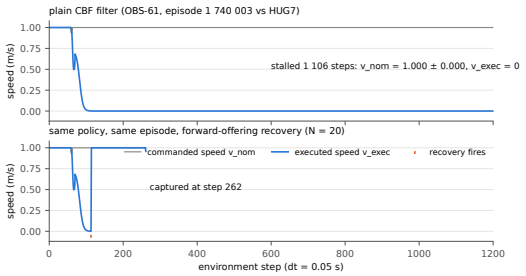


Figure 1: Frozen input: nominal and executed speed over one archived episode under the plain filter (round S17) and the same seed under the recovery, which captures after one firing.

Proposition 3 (Recovery) *(a) Every command the recovery executes is an output of the base filter on the current state, so the forward-invariance guarantee is exactly the base filter's. (b) If the stalled position is not a position trap, some heading admits forward motion; the admissible rotation reaches it within $T_{\mathrm{turn}} = 45$ steps at $\omega_{\max}$, after which the forward candidate passes, so the body typically leaves the position within $N + T_{\mathrm{turn}}$ steps. (c) Nothing in (a)–(b) excludes a stall–recover–stall cycle; it is measured, not bounded.*

Part (b) says "typically" because round B2 found 97.8% of 10 475 triggers within the bound; the tail is reported, not explained. Any filter that executes the nearest admissible command, including the closest-safe-action shield of Kim et al. (2025), satisfies the hypothesis of Proposition 1; the recovery is the minimal state that breaks (R2) without changing $\mathcal{A}(s)$.

5. Experiments

Protocol. Each round was specified before any number existed: hypotheses, bars, verdict rules, and the sentence to write if a hypothesis failed. Code was frozen at a commit, cells used the same seeds across arms on fresh streams, and paired differences were tested with the exact two-sided binomial test on discordant pairs at $p < 0.05$. Construct gates refuse to summarise a round until shared cells reproduce the earlier records bit for bit and designated animated episodes have been watched; a stop rule forbids re-running or re-cutting a cell after its number exists. Every verdict below is the registered one, including the failures.

5.1. Phenomenon and mechanism

The filter's cost is large before any remedy: in round S4 the learned pursuer STR14 captured the greedy evader in 0.728 of 500 paired episodes without the filter and 0.396 with it ($p = 3 \times 10^{-44}$), and 0.540 of the filtered episodes were deadlock episodes. In round S5 a stall penalty alone recovered nothing (0.426, 0.464, 0.436 against 0.426) while filter-state features recovered part of the loss on three of three seeds (0.522, 0.512, 0.544; $p \leq 8 \times 10^{-4}$) and left 0.11–0.30 of episodes in deadlock.

Round S15 measured the mechanism on 772 stall entries (Figure 2). On the pre-registered hold-10 cells (383 entries, 97 escaped) an escaped stall is followed by a capture in 0.299 of cases against 0.042 otherwise, with a mean delay of 629 steps; the discounted return-to-go differs by 0.0092. During a stall the world observation does not change (median change 0.000) and the evader

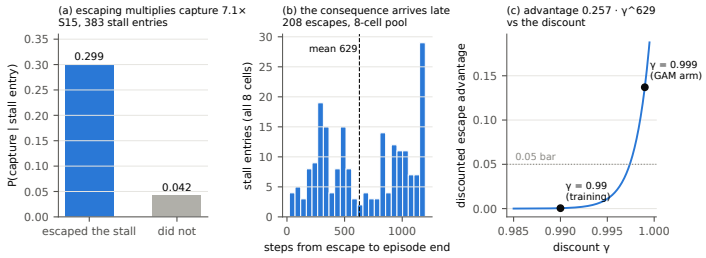


Figure 2: Mechanism (round S15). (a) Capture rate after an escaped stall versus not. (b) Delay from escape to capture. (c) The bound $\gamma^T A_0$ against γ.

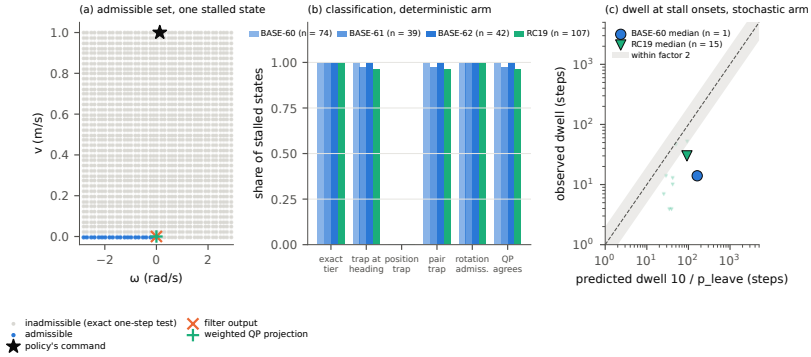


Figure 3: Stall geometry (round B1). (a) Admissible set at one archived stalled state with the policy's command, the filter's output and the exact projection. (b) Classification of stalled states, deterministic arm. (c) Observed against predicted dwell, stochastic arm, factor-two band.

is visible in 2.4% of stalled steps; reducing the decision hold from 10 steps to 1 changes immobility by less than 0.05. The input is frozen and the escape's reward lies beyond the training horizon.

Round B1 measured Proposition 1 on four policies against the learned attacker, 100 episodes each with deterministic actions and the same seeds with stochastic sampling (Figure 3; per-policy shares in the supplement). All 262 stalled states lie on the exact tier; 98.1% are traps at the current heading, none is a position trap, and an admissible rotation exists at every one. The exact weighted projection agrees with the filter's output in 98.1% of states and yields forward motion in the remaining 1.9%, the five states that are not traps at heading; the registered bar was 99%, so Proposition 1 is stated with this share: in 98% of stalled states the stall is the admissible set's, in 2% the filter's enumeration's. The registered dwell test failed: deadlock under stochastic sampling is 0.42, 0.24, 0.29, 0.57 against 0.38, 0.20, 0.21, 0.55 deterministic, so the loop survives exploration on four of four policies, but the geometric prediction could not be tested on two policies, whose stochastic-arm onsets all occurred on the intercept-law branch, and over-predicted the observed median dwell 3- to 12-fold on the other two. The intercept law produced 57 to 100% of the stalled states; the same stall was seen in a rule-based pursuer in a separate game. Self-stops are 41 to 66% of the BASE policies' immobile steps and absent for RC19.

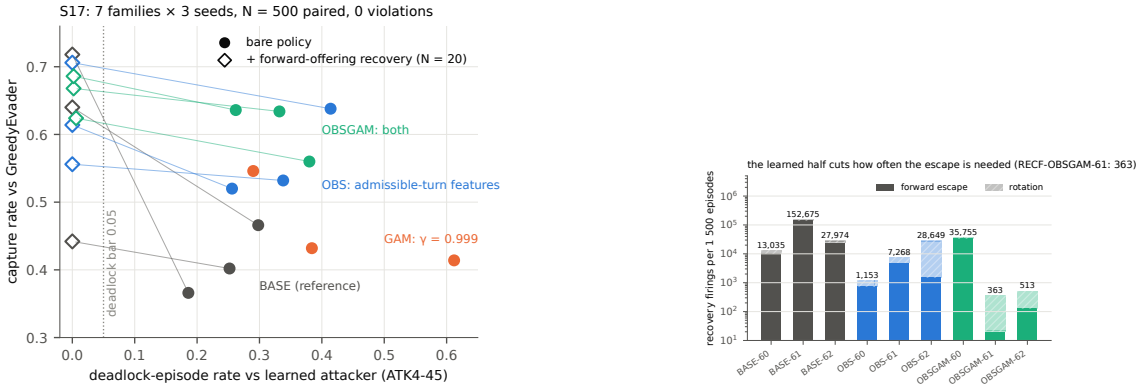


Figure 4: Round S17, $N = 500$ paired. Left: deadlock-episode rate against the learned attacker versus capture against the greedy evader, 21 arms; hue marks the family, shape the presence of the recovery; the registered bar is 0.05. Right: recovery firings per family over 1 500 gate episodes, forward versus rotation candidates.

5.2. Remedies

Round S16 trained five learning-side remedies from scratch, three seeds each, plus a scripted rotation-only recovery. Observation features and a longer horizon lowered immobility and raised capture without clearing the deadlock; the recurrent arm was confounded by an amendment to its batch settings and licenses no conclusion about memory; the stall penalty backfired (Section 6). The rotation-only recovery scored 0.000 deadlock episodes in all nine cells by spinning, up to 277 firings per episode, although a straight command would have moved the body in every sampled stalled pose; hence forward candidates come first.

Round S17 compared seven families of three seeds (Figure 4). Against the learned attacker the BASE family's deadlock-episode rate is 0.298, 0.186, 0.252; with the forward-offering recovery at $N = 20$ it is 0.000 on the three BASE and three OBS seeds and 0.006, 0.002, 0.002 on the three OBSGAM seeds, against a registered bar of 0.05. Capture against the greedy evader rises significantly on nine of nine recovery seeds (BASE 0.466, 0.366, 0.402 to 0.640, 0.718, 0.442; $p \leq 3.6 \times 10^{-5}$); the bare learned families raise capture too (OBS 0.532, 0.520, 0.638; GAM 0.546, 0.432, 0.414; OBSGAM 0.560, 0.634, 0.636) and leave deadlock between 0.26 and 0.61. Every remedy that learns improves the task; only the escape removes the failure. Learning changes how often the escape is needed: the RECF-BASE seeds fire 13 035, 152 675 and 27 974 times over 1 500 episodes, the RECF-OBSGAM seeds 35 755, 363 and 513, for equal capture. A registered anti-spin test passed on every arm and is uninformative, since the arm it was written to catch was not in the round.

Round B2 swept $N \in \{5, 10, 20, 40, 80\}$ on five policies and added a perturbation baseline (Figure 5). Capture against the greedy evader is significantly above the plain filter at every N on all five policies and never significantly below it; deadlock against the attacker is at most 0.022 for $N \leq 20$ on all five (BASE 0.000; STR14 0.008, 0.022, 0.014; RC19 0.000, 0.000, 0.002) against 0.186 to 0.710 plain. For $N \geq 40$ the 40-step indicator is insensitive by construction; runs of at least 80 steps are 0.000 to 0.020 of episodes at $N = 40$ and runs of at least 160 steps 0.000 to 0.016 at $N = 80$, against 0.176 to 0.682 plain. For the policies of the discovery, STR14's capture rises

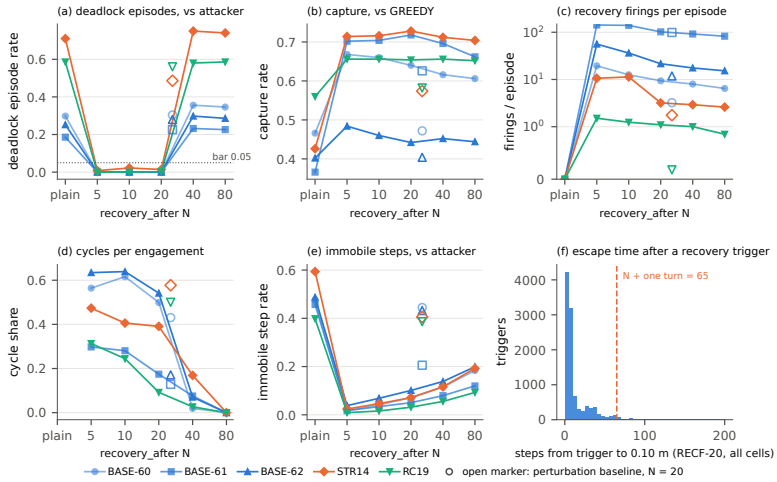


Figure 5: Round B2, five policies, $N = 500$ paired, against the recovery parameter: (a) deadlock-episode rate (attacker), (b) capture (greedy evader), (c) firings per episode, (d) cycles per engagement, (e) immobile-step rate (attacker); open markers are the perturbation baseline at $N = 20$. (f) Steps from trigger to a $0.10\,\mathrm{m}$ displacement at $N = 20$, with the $N + T_{\mathrm{turn}}$ bound.

from 0.426 to 0.728 ($p = 7 \times 10^{-46}$) and its immobile-step rate falls from 0.594 to 0.071; RC19's rise from 0.560 to 0.654 ($p = 10^{-14}$) and fall from 0.398 to 0.032.

The bound of Proposition 3(b) holds for 97.8% of the 10 475 decidable triggers at $N = 20$; eight of ten cells exceed 98.5%, and BASE-62's two cells sit at 94.5 and 92.3%, below the registered 95%. The median escape takes 4 to 8 steps at every N. The cost is cycles, not immobility: at $N = 20$ the immobile-step rate is 0.023 to 0.101 on every cell and 0.08 to 0.56 of engagements are cycles.

The perturbation baseline draws up to eight commands uniformly from the box after 20 immobile steps, filters each, and executes the first whose filtered speed exceeds $0.05\,v_{\max}$: the usual random resolution of a deadlock, registered with the expectation that it would clear the deadlock and lose on capture and firings. It clears the deadlock on none of the five policies (0.304, 0.226, 0.280, 0.486, 0.560 against 0.298, 0.186, 0.252, 0.710, 0.586 plain); the recovery captures more on five of five and fires more on five of five. Figure 3(a) explains both: at a trap at heading the admissible set is the zero-speed segment, no sampled command moves, and a baseline that executes only moving commands can never rotate out; its firings land on self-stops and never on the trap (one policy fires it 99 times per episode with deadlock unchanged), whereas 5 to 89% of the recovery's firings at $N = 20$ are rotations. The recovery's contribution is the rotation candidate, not the trigger.

5.3. A second filter

Round B3 replaced the CBF projection with a sampling lookahead of horizon 4 (Figure 6). Deadlock against the attacker is 0.104, 0.036, 0.046, 0.468 on the four policies against 0.298, 0.186, 0.252, 0.586 under the CBF filter; paired, the CBF filter deadlocks more on seven of eight cells ($p \leq 0.002$), the exception being RC19 against the greedy evader (0.386 against 0.284). The registered bar, at least 0.05 on three of four policies, was met on two: the loop is present but weaker under

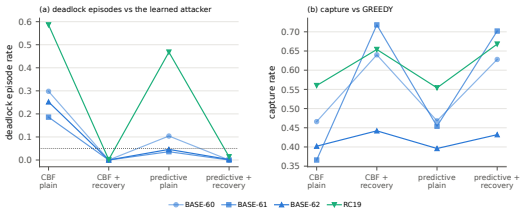


Figure 6: Rounds B3 and B4, four policies, $N = 500$ paired: (a) deadlock-episode rate against the attacker and (b) capture against the greedy evader under the CBF and predictive filters, without and with the recovery at $N = 20$.

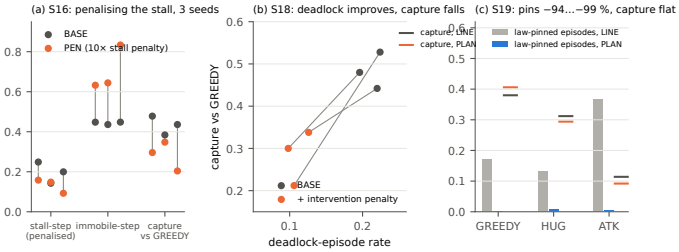


Figure 7: Three pre-registered dissociations. (a) Round S16: the stall penalty against its baseline, three seeds. (b) Round S18: the correction-magnitude penalty, three seeds. (c) Round S19: episodes with the law held at a stall, and capture, line law versus planner law, three opponents.

the predictive filter at this lookahead. Round B4 composed the recovery onto the predictive filter so that every candidate is accepted or rejected by the predictive test (a construct gate re-filtered all 39 162 commands of the first five episodes of each cell without a rejection). The composition clears the deadlock (0.000, 0.000, 0.002, 0.014) and raises capture over the plain predictive filter on four of four policies (0.468 to 0.628, 0.454 to 0.702, 0.396 to 0.432, 0.554 to 0.668; all $p < 0.05$).

6. Three dissociations: liveness metrics versus the task

Each of the following pre-registered hypotheses failed in the registered direction (Figure 7).

A stall penalty teaches the agent to stop asking. In round S16 the PEN family, trained with a stall penalty, halves the stalled-step rate (0.143–0.249 to 0.093–0.158) while its immobile-step rate rises from 0.436–0.448 to 0.633–0.833 and its capture against the greedy evader falls from 0.384–0.478 to 0.204–0.348. The policy learns to emit a zero nominal (R4); the fixed point is the same.

A correction-magnitude penalty improves the deadlock metric and hurts the task. Round S18 trained 18 arms with the penalty of Pizarro Bejarano et al. (2025) on the magnitude of the filter's correction (they square it; we use the unsquared norm in normalised units, calibrated to an expected episode penalty of one). In their quadrotor tracking task the penalty raised return and reduced chattering; standing still is never a local optimum there. In the dense-reward column capture falls

on three of three seeds (0.442, 0.480, 0.528 to 0.338, 0.300, 0.212; pooled $p = 3.6 \times 10^{-51}$) while zero-motion requests rise from 0.07, 0.16, 0.00 to 0.37, 0.78, 0.80, and the pre-registered deadlock metric improves in nine of nine cells (one seed from 0.224 to 0.106 while its immobile-step rate rises from 0.23 to 0.81). The penalty was also charged for the intercept law's interventions, a registered confound.

Removing the pins removes a symptom. Round S19 replaced the intercept law with a planner that avoids obstacles. Episodes with the law held at a stall for at least 40 steps fall from 0.170, 0.132, 0.368 to 0.002, 0.008, 0.004 against the three opponents, while filtered capture changes by $+0.026$, -0.018, -0.022 ($p = 0.015, 0.035, 0.007$). The pins were a symptom.

Consequence. Evaluations of filtered learned policies should report the task, immobility and stalled steps separately, and firings per episode for any recovery; a liveness metric that improves under a penalty is a warning, not a result, until the task moves with it.

7. Discussion and limitations

The claims are about memoryless policies; the one recurrent arm was confounded before its number existed, so memory is untested. One game, one dynamics model, two filter constructions; three seeds per family, 500 paired episodes per cell. Remark 2 is a bound: the longer-horizon family shows the horizon is implicated and lengthening it insufficient. The intercept law's stalls are the same fixed point as the learned head's.

The recovery has a cost: cycles are 8 to 56% of engagements at $N = 20$, the escape bound fails on 2.2% of triggers, and the firing count varies by three orders of magnitude across policies that capture equally well. A random perturbation that accepts only moving commands cannot leave a trap at heading; one that accepted rotations would be a different baseline, not run here.

Fifteen specifications passed their bars while measuring the artefact this paper is about; the failure was found by a person watching the animated episodes (1 085 consecutive blocked requests, command standard deviation 0.002, an unregistered diagnostic of 100 episodes). Watching the behaviour is now a construct gate in our protocol. In practice: keep the filter, add the counter, log the firings, watch the episodes.

8. Conclusion

A minimal-intervention safety filter in front of a memoryless learned policy can create a closed-loop fixed point at which the guarantee holds, violations are zero, and the task fails silently. We characterised it, showed that its geometry is the admissible set's, measured why the learner cannot see the way out, and showed across seven pre-registered rounds that learning-side remedies improve the task without removing the failure while a one-counter recovery from the same admissible set removes it with the guarantee untouched. The failure is cheap to remove and easy to miss; the second point is the one we would like this paper to change.

Reproducibility. Environment, filters, policies, pre-registrations, frozen commits, per-cell records and animated episodes for every round are released at [repository URL]; each record names its commit and seed stream.

Generative AI Use

Generative AI tools were used for methodology drafting, coding and experimental implementation, literature search, and writing and editing. The authors reviewed and verified all AI-assisted content and take responsibility for the final content of the paper.

References

Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 2669–2678, 2018. doi: 10.1609/aaai.v32i1.11797.

Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *Proceedings of the 18th European Control Conference (ECC)*, pages 3420–3431, 2019. doi: 10.23919/ECC.2019.8796030.

Daniel Arnström and André M. H. Teixeira. Data-driven and stealthy deactivation of safety filters. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference (L4DC)*, volume 283 of *Proceedings of Machine Learning Research*, pages 456–468, 2025.

Richard Cheng, Gábor Orosz, Richard M. Murray, and Joel W. Burdick. End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control tasks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 3387–3395, 2019. doi: 10.1609/aaai.v33i01.33013387.

Murad Dawood, Ahmed Shokry, and Maren Bennewitz. A dynamic safety shield for safe and efficient reinforcement learning of navigation tasks. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference (L4DC)*, volume 283 of *Proceedings of Machine Learning Research*, pages 686–697, 2025.

Tiberiu Georgescu, Tom Goodall, Dalal Alrajeh, Francesco Belardinelli, and Sebastian Uchitel. Adaptive GR(1) specification repair for liveness-preserving shielding in reinforcement learning. In *Proceedings of the International Conference on Neuro-symbolic Systems (NeSy)*, 2026. arXiv:2511.02605.

Jaskaran Grover, Changliu Liu, and Katia Sycara. Why does symmetry cause deadlocks? *IFAC-PapersOnLine*, 53(2):9746–9753, 2020. doi: 10.1016/j.ifacol.2020.12.2644.

Jaskaran Grover, Changliu Liu, and Katia Sycara. Deadlock analysis and resolution for multi-robot systems. In *Algorithmic Foundations of Robotics XIV (WAFR 2020), Springer Proceedings in Advanced Robotics*, pages 294–312, 2021. doi: 10.1007/978-3-030-66723-8_18.

Jaskaran Grover, Changliu Liu, and Katia Sycara. The before, during, and after of multi-robot deadlock. *The International Journal of Robotics Research*, 42(6):317–336, 2023. doi: 10.1177/02783649221074718.

Kai-Chieh Hsu, Haimin Hu, and Jaime F. Fisac. The safety filter: A unified view of safety-critical control in autonomous systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 7(1):47–72, 2024. doi: 10.1146/annurev-control-071723-102940.

Kyungmin Kim, Davide Corsi, Andoni Rodríguez, Nicholas Lanier, Benjami Parellada, Pierre Baldi, César Sánchez, and Roy Fox. Realizable continuous-space shields for safe reinforcement learning. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference (L4DC)*, volume 283 of *Proceedings of Machine Learning Research*, pages 932–945, 2025.

Donggeon David Oh, Duy P. Nguyen, Haimin Hu, and Jaime F. Fisac. Provably optimal reinforcement learning under safety filtering. In *Proceedings of the International Association for Safe and Ethical AI Conference (IASEAI'26)*, volume 2, pages 488–500, 2026. arXiv:2510.18082.

Federico Pizarro Bejarano, Lukas Brunke, and Angela P. Schoellig. Safety filtering while training: Improving the performance and sample efficiency of reinforcement learning agents. *IEEE Robotics and Automation Letters*, 10(1):788–795, 2025. doi: 10.1109/LRA.2024.3512374.

Matheus F. Reis and A. Pedro Aguiar. On the stability of undesirable equilibria in the quadratic program framework for safety-critical control. *Automatica*, 190:113032, 2026. doi: 10.1016/j.automatica.2026.113032.

Matheus F. Reis, A. Pedro Aguiar, and Paulo Tabuada. Control barrier function-based quadratic programs introduce undesirable asymptotically stable equilibria. *IEEE Control Systems Letters*, 5(2):731–736, 2021. doi: 10.1109/LCSYS.2020.3004797.

Matheus F. Reis, Gonçalo Carvalho, and A. Pedro Aguiar. A unified stability analysis of safety-critical control using multiple control barrier functions. arXiv:2503.15823, 2025.

Xiao Tan and Dimos V. Dimarogonas. On the undesired equilibria induced by control barrier function based quadratic programs. *Automatica*, 159:111359, 2024. doi: 10.1016/j.automatica.2023.111359.

Yuhao Wang, Xiaoyu He, and Miroslav Pajic. Neuro-symbolic deadlock resolution in multi-robot systems. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference (L4DC)*, volume 283 of *Proceedings of Machine Learning Research*, pages 1066–1077, 2025.

Yuxuan Wu, Zhaoyang Yang, Donggeon David Oh, Duy P. Nguyen, and Haimin Hu. Turning safety into competence: Minimally exploitable robot policies via safety-filtered reinforcement learning. arXiv:2609.27312, 2026.